

EduVision DV

Higher Education Performance Dashboard

Final Project Documentation

Covers data sources, data model, KPI methodology, all four analytical dashboards, testing and quality assurance, and project delivery.

Prepared by: Vepanjeri Hasini

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1. Executive Summary

EduVision DV is a higher education analytics project built around a single Tableau workbook containing four analytical dashboards: University Overview, Research Analytics, Student Analytics, and Country Comparison. The workbook is fed by one analytical dataset assembled from four public sources: QS World University Rankings 2025, Times Higher Education (THE) World University Rankings 2024, World University Rankings 2023 (WUR), and World Bank Education Statistics (EdStats).

This document is the consolidated, final record of the project. It explains what the data is and how it was prepared, what each KPI means and how it is calculated, how each dashboard is built and used, what was tested and the results, and how the finished project is organised and delivered.

1.1 Headline Results

- Four public source datasets cleaned, standardised, and merged into one star-schema analytical model (seven sheets in `eduvision_final_dataset.xlsx`).
- **4,708 universities** in the university dimension, with **6,409 university, source, and year records**: QS 2025 (1,503), THE 2024 (2,673), and WUR 2023 (2,233).
- **136 countries** in the country dimension, of which **119** have World Bank education indicators.
- Six core KPIs engineered in Python and validated: Global Ranking Score, Research Impact Score, Faculty-to-Student Ratio, International Student Percentage, Academic Reputation Score, and Research Productivity Index.
- Four dashboards built and linked through left-hand navigation buttons, with Year, Region, University, and Country filters where applicable.
- **30 QA test cases** recorded: 26 PASS, 2 PASS WITH NOTE, and 2 flagged for final manual confirmation. All 14 KPI card values were independently recomputed from the Excel model and matched after rounding.
- Referential integrity confirmed: 0 duplicate university IDs, 0 duplicate country IDs, 0 invalid university foreign keys, and 0 invalid country foreign keys.

1.2 Document Scope

This document covers every task listed under Milestone 4 (Testing and Delivery) and consolidates the outputs of Milestones 1 to 3, so a new reader does not need to open the earlier milestone files to understand the finished product. A reviewer with no prior context should be able to read this document, open the Tableau workbook, and understand what they are looking at.

2. Project Overview

2.1 Objective

The project develops a higher education analytics dashboard suite for analysing university rankings, research performance, student diversity, academic excellence, and global education trends, using publicly available ranking and education statistics. The final deliverable is a Tableau workbook with four linked dashboards, supported by a documented and validated data model.

2.2 Audience

The dashboards are designed for users with different goals: students comparing universities, researchers benchmarking research output and citation impact, university administrators tracking institutional position, policymakers comparing country-level education indicators, and education consultants preparing comparisons for clients.

2.3 The Four Dashboards

#	Dashboard	Role	Primary Grain
1	University Overview	Global ranking, headline KPIs, regional spread, academic reputation	One row per university, source, and year
2	Research Analytics	Research score, citation score, research impact and productivity	One row per university, source, and year
3	Student Analytics	Enrollment, international share, faculty-to-student ratio	One row per university, source, and year
4	Country Comparison	Country-level literacy and tertiary enrollment indicators	One row per country

2.4 Project Timeline

The project was organised into four milestones, each made up of two modules:

1. **Milestone 1:** Data Collection and Preparation (Module 1 data loading, Module 2 cleaning and standardisation).
2. **Milestone 2:** KPI Engineering and Dashboard Planning (Module 3 star data model, KPIs and validation; Module 4 dashboard storyboard and Tableau prototype).
3. **Milestone 3:** Dashboard Development (Module 5 University Overview and Research Analytics; Module 6 Student Analytics, Country Comparison, and integration).
4. **Milestone 4:** Testing and Delivery (Module 7 testing and validation; Module 8 documentation and project delivery). This document is the primary Module 8 deliverable.

3. Data Sources and Data Model

3.1 Source Datasets

Source	Provider	Records in Model	Used For
QS World University Rankings 2025	Quacquarelli Symonds	1,503 universities (year 2025)	Global rank, overall score, academic and employer reputation, international and faculty-student scores
THE World University Rankings 2024	Times Higher Education	2,673 universities (year 2024)	Rank, overall score, research, citations, student numbers, student-staff ratio, international %
World University Rankings 2023	Aggregated ranking release	2,233 universities (year 2023)	Rank, overall score, research, citations, student numbers, student-staff ratio, international %
World Bank Education Statistics (EdStats)	World Bank	119 countries with indicators	Country comparison: literacy, government expenditure, enrollment ratios

3.2 Star Data Model

The cleaned sources are organised as a star schema. Two dimension tables describe universities and countries, and four fact tables hold measurements. The same tables are exported as CSV files and as sheets in `eduvision_final_dataset.xlsx`, which is the single file the Tableau workbook is built on.

Table	Grain / Key Fields	Rows	Role
dim_university	university_id, display_name, region, qs_name, the_name, wur_name, in_qs, in_the, in_wur, country_id, country_name	4,708	One row per unique university across all three ranking sources
dim_country	country_id, country_name	136	One row per country
fact_university_performance	university_id + year + source; global_rank, overall_score, academic_reputation, employer_reputation, global_ranking_score	6,409	Rank and score per university per source
fact_research	university_id + year + source; research_score, citation_score, research_impact, research_productivity_index	6,409	Research and citation indicators
fact_student	university_id + year + source; total_students, students_per_staff, international_student_percentage, QS_scores, faculty_student_ratio	6,409	Student body indicators

Table	Grain / Key Fields	Rows	Role
fact_country_education	country_id; adult literacy, government expenditure (% of GDP and % of government spending), primary and tertiary enrollment ratios	119	World Bank indicators, one column per indicator
kpi_dataset	university_id + year + source; all six KPIs plus university and country attributes (20 columns)	6,409	Flat analytical table used by the dashboards

3.3 Relationship Pattern

- dim_university is the hub. Each fact table links to it through **university_id**.
- dim_university links to dim_country through **country_id**, and fact_country_education links to dim_country through **country_id**.
- Because one university can appear in several sources and years, **university_id alone is not a unique row key**. The unique key for the fact tables is the composite **university_id + year + source**.

3.4 Data Cleaning and Transformation

The data preparation was carried out in Jupyter notebooks and followed these stages:

1. **Data loading (01_data_loading.ipynb)**: the four raw files (qs_2025_raw.csv, the_2024_raw.csv, wur_2023_raw.csv, world_bank_education_subset.csv) were loaded and profiled.
2. **Cleaning (02_data_cleaning.ipynb)**: each source was cleaned and saved as a *_clean.csv file.
3. **Standardisation and matching (03_data_standardization.ipynb)**: university and country names were standardised so the same institution or country is represented consistently across sources, and dim_university and dim_country were built. Matches that needed human judgement were written to matches_for_manual_review.csv, and World Bank country matches to world_bank_matched.csv. Unmatched countries were logged rather than guessed into the model.
4. **KPI engineering (04_kpi_engineering.ipynb)**: the fact tables and the six KPIs were built (Section 4).
5. **Validation (05_validation.ipynb)**: identifiers, keys, and referential integrity were checked (Section 11).

3.5 Missing Values

Missing values were never filled with zero. Where a source does not publish an indicator, the cell is left blank so that "not reported" is not mistaken for a score of zero. The share of blank values in the main kpi_dataset columns is shown below.

Column	Missing	Reason
global_rank	20.4%	Not every ranked row has a numeric rank

Column	Missing	Reason
global_ranking_score / overall_score	84.4%	QS and THE publish an exact overall score only for their top ranking tier
academic_reputation_score, employer_reputation	76.5%	Published by QS only
research_impact_score, research_productivity_index	43.4%	Needs both research score and citation score; QS has no direct research score
faculty_student_ratio, total_students	23.8%	QS publishes a score instead of the raw ratio and student count
international_student_percentage	23.9%	QS publishes a score, not a percentage
region	50.0% of rows (3,205 of 4,708 universities)	No region was assigned to these universities in the source data
country_id / country_name	2.9% of rows (186 universities)	Country could not be matched to the reference list

3.6 Referential Integrity Result

The final model passed referential integrity validation: 0 duplicate university IDs, 0 duplicate country IDs, 0 duplicate composite keys in the fact tables, 0 invalid university foreign keys, and 0 invalid non-null country foreign keys.

4. KPI Definitions and Methodology

Six KPIs were engineered in Python (pandas) in 04_kpi_engineering.ipynb before loading into Tableau. Doing this in Python keeps the Tableau workbook fast and keeps the KPI logic transparent and reproducible. Each KPI below lists its definition, source fields, and where it appears.

4.1 Global Ranking Score

Definition: the overall_score of the ranking source a row comes from (QS 2025, THE 2024, or WUR 2023), rescaled to 0 to 100 using min-max scaling within each source and year. Rescaling is needed because the sources do not publish scores on directly comparable scales. Where the source overall score is missing, the value is left blank.

Derived from: overall_score, global_rank. **Appears in:** University Overview (Global Rank, Overall Score, Top Universities, Performance Trend).

4.2 Research Impact Score

Definition: the mean of research_score and citation_score. It is calculated only where both inputs exist, so a missing input is never treated as zero. For QS rows, citations per faculty is used as the closest documented proxy for citation score and no research score is available.

Derived from: research_score, citation_score. **Appears in:** Research Analytics (Research Impact, Research Impact Comparison) and University Overview (Research Impact).

4.3 Faculty-to-Student Ratio

Definition: students per staff member, taken from the students_per_staff field of THE 2024 and WUR 2023. A lower value generally means more individual attention. QS publishes a 0 to 100 faculty-student score instead of a ratio; it is stored separately as faculty_student_score_qs and is never presented as the ratio.

Derived from: students_per_staff. **Appears in:** Student Analytics (Faculty Student Ratio, Faculty-to-Student Ratio Analysis) and University Overview.

4.4 International Student Percentage

Definition: the share of enrolled students who are international, taken from THE 2024 and WUR 2023, which report a true percentage. QS reports a 0 to 100 score, which is stored separately as international_student_score_qs. The flag international_student_is_true_percentage marks which rows are true percentages so the two are never mixed.

Derived from: international_student_percentage. **Appears in:** Student Analytics (International Students, International Student Analysis, Student Diversity Trends) and University Overview.

4.5 Academic Reputation Score

Definition: the QS academic reputation survey score, a measure of how highly academic peers regard an institution. QS is the only one of the three ranking sources that publishes it, so the KPI is populated for QS 2025 rows only. Employer reputation is stored alongside it.

Derived from: academic_reputation, employer_reputation. **Appears in:** University Overview (Academic Reputation, UO-Academic Reputation chart).

4.6 Research Productivity Index

Definition: the percentile rank (0 to 100) of research_impact within each source and year. It shows where an institution sits relative to others in the same ranking, on a scale that is fair across sources.

Derived from: research_impact. **Appears in:** Research Analytics (Research Productivity KPI and Research Productivity Trends).

4.7 KPI Summary Table

KPI	Unit	Dashboard Value
Global Rank (minimum rank)	Rank	1.000
Average Overall Score	Score	52.50
Average Academic Reputation	Score 0 to 100	20.29
Average Research Impact	Score 0 to 100	37.22
Average Research Productivity	Index 0 to 100	50.03
Average Faculty-Student Ratio	Students per staff	19.07
Average International Student %	Percent	9.95%

5. Dashboard Architecture

5.1 Navigation Model

Every dashboard carries the same left-hand navigation panel with four buttons: University Overview, Research Analytics, Student Analytics, and Country Comparison. A user can reach any dashboard from any other with one click.

5.2 Filters by Dashboard

Dashboard	Filters
University Overview	Year (range slider, 2023 to 2025), Region (multi-select dropdown)
Research Analytics	Year (range slider, 2023 to 2025), Display Name (university), Country Name
Student Analytics	Year (range slider, 2023 to 2025), Display Name (university)
Country Comparison	Country Name (dropdown), Region (multi-select dropdown and colour legend)

5.3 Visual Design Standard

All four dashboards share one layout: a pale green background, the "Edu Vision DV" title with the dashboard name and subtitle "Higher Education Performance Analytics" at the top, a row of KPI cards beneath it, the navigation buttons on the left, and a grid of charts. Worksheets use a consistent prefix so they are easy to find: UO- (University Overview), RA- (Research Analytics), SA- (Student Analytics), and CC- (Country Comparison), with a "-KPI" suffix for headline cards.

6. Dashboard Guide: University Overview

University Overview is the starting point for ranking questions: who is ranked where, how universities are spread across regions, how scores move over time, and which institutions lead on academic reputation.

6.1 Worksheets on This Dashboard

Worksheet	Chart Type	What It Shows
UO-KPI Global Rank	Single value card	Best (minimum) global rank in the current filter context, 1.000
UO-KPI Overall Score	Single value card	Average overall score, 52.50
UO-KPI Academic Reputation	Single value card	Average academic reputation score, 20.29
UO-KPI Research Impact	Single value card	Average research impact score, 37.22
UO-KPI Faculty Student Ratio	Single value card	Average students per staff, 19.07
UO-KPI International Students	Single value card	Average international student percentage, 9.95%
UO-Top Universities	Horizontal bar	Universities ordered by best global rank, with the top-ranked institution first
UO-Universities by Region	Horizontal bar	Distinct count of universities in each world region, including the unassigned (Null) group
UO-Performance Trend	Line chart	Average global ranking score by year across 2023 to 2025
UO-Academic Reputation	Horizontal bar	Universities with the highest average academic reputation score

6.2 Usage Notes

- Use the Year slider to narrow the view to one ranking year or a range; the KPI cards and charts respond to it.
- Use the Region filter to focus on one or more world regions. A large group of universities has no region in the source data and appears as Null (Section 12).
- Academic reputation is published by QS only, so the reputation chart reflects QS 2025 records.

7. Dashboard Guide: Research Analytics

Research Analytics answers questions about research output: which institutions score highest on research, how citation strength relates to research score, and how research impact differs by region.

7.1 Worksheets on This Dashboard

Worksheet	Chart Type	What It Shows
RA-KPI Research Score	Single value card	Average research score, 23.38
RA-KPI Citation Score	Single value card	Average citation score, 42.98
RA-KPI Research Impact	Single value card	Average research impact, 37.22
RA-KPI Research Productivity	Single value card	Average research productivity index, 50.03
RA-Research Impact Comparison	Horizontal bar	Average research impact by world region
RA-Top Research Institutions	Horizontal bar	Average research score by university, coloured by region
RA-Research Score Analysis	Horizontal bar	Average research score by university
RA-Research Productivity Trends	Line chart	Average research productivity index by year (data exists for 2023 and 2024)
RA-Citation Performance	Scatter plot	Average citation score against average research score, one point per university, coloured by region

7.2 Usage Notes

- The scatter plot is the quickest way to spot institutions that are strong in research but weaker in citations, or the reverse.
- The Region legend colours the university-level charts consistently, so a region can be followed across the dashboard.
- Research score is available for THE 2024 and WUR 2023 only. QS 2025 does not publish a research score, so research metrics do not cover 2025.

8. Dashboard Guide: Student Analytics

Student Analytics answers questions about the student body: how large institutions are, how international they are, and how many students there are per faculty member.

8.1 Worksheets on This Dashboard

Worksheet	Chart Type	What It Shows
SA-KPI Total Students	Single value card	Total enrolled students in the current filter context, 92.41M
SA-KPI International Students %	Single value card	Average international student percentage, 9.95%
SA-KPI Faculty Student Ratio	Single value card	Average students per staff, 19.07
SA-KPI International Student Score	Single value card	Average QS international student score, 25.58
SA-KPI Faculty Student Score	Single value card	Average QS faculty-student score, 28.13
SA-Student Diversity Trends	Stacked bar	Domestic student % and true international student % for each university shown
SA-International Student Analysis	Horizontal bar	Average true international student % by university
SA-Faculty-to-Student Ratio Analysis	Horizontal bar	Average faculty-student ratio by university
SA-Enrollment Comparisons	Horizontal bar	Total students by university
SA-Student Distribution Analysis	Scatter plot	Total students against true international student %, one point per university

8.2 Usage Notes

- The international percentage charts use only rows where the source reports a true percentage (THE 2024 and WUR 2023). QS publishes a score instead, which is shown separately in the International Student Score card and never relabelled as a percentage.
- Total Students is the sum of all records in the selected Year range. A university can appear in more than one source and year, so select a single year to avoid counting the same institution more than once.
- Faculty-to-student ratio is displayed as students per staff (for example 19.07).

9. Dashboard Guide: Country Comparison

Country Comparison answers questions at the national level: how countries compare on adult literacy and tertiary enrollment, and how these two indicators relate to each other.

9.1 Worksheets on This Dashboard

Worksheet	Chart Type	What It Shows
CC-KPI Total Countries	Single value card	Distinct countries in the model, 136
CC-KPI Education Countries	Single value card	Countries with World Bank education indicators, 119
CC-KPI Avg Literacy	Single value card	Average adult literacy rate (%), 87.97
CC-KPI Avg Tertiary Enrollment	Single value card	Average tertiary enrollment ratio (%), 44.33
CC-Regional Education Comparison	Horizontal bar	Average tertiary enrollment ratio by country, coloured by region
CC-Country Ranking Comparison	Horizontal bar	Countries by minimum global ranking score of their universities
CC-Education Performance Benchmarking	Scatter plot with trend lines	Average adult literacy against average tertiary enrollment, coloured by region
CC-Top Performing Countries	Horizontal bar	Countries with the highest average adult literacy rate

9.2 Usage Notes

- The Country Name dropdown narrows every chart to one country; the Region control isolates one or more regions.
- The scatter plot shows the relationship between literacy and tertiary enrollment, with a trend line per region.
- Enrollment ratios can exceed 100% in World Bank data because they include students outside the official age group.
- World Bank data is held as one row per country, so this dashboard shows country averages rather than a year-by-year trend.

10. Filters, Navigation, and Dashboard Interactions

10.1 Navigation

The four navigation buttons on the left of each dashboard open the other dashboards. Each button was manually tested in Tableau by the project owner (QA-022).

10.2 Filters

Year, Region, University (Display Name), and Country Name filters update the views on the dashboards where they appear (QA-023 and QA-024). The filters are limited to what each dashboard needs so that a first-time user is not overwhelmed.

10.3 Dashboard Linking

Dashboard linking was manually checked by the project owner and recorded as working in the Dashboard Testing Report. The QA Checklist lists QA-025 (dashboard actions) with a suggested manual path of selecting one university or country.

11. Testing and Quality Assurance Summary

Full detail lives in two companion documents delivered with this one: the **Milestone 4 QA Checklist** and the **Milestone 4 Dashboard Testing Report**. This section summarises their outcome.

11.1 What Was Tested

- **Data integrity:** uniqueness of university and country IDs, the composite key of the fact tables, foreign keys, and missing-value documentation.
- **KPI calculations:** every KPI card value was independently recomputed from the final Excel model and compared with the value displayed in Tableau.
- **Ranking:** the displayed global rank of 1 was checked against the minimum global_rank in the performance fact table.
- **Educational metrics:** average literacy and tertiary enrollment were checked against fact_country_education.
- **Dashboard behaviour:** presence of all four dashboards, navigation, Year filter, University and Country filters, linking, and visual quality.

11.2 Results by Area

Area	Test IDs	Cases	Result
Data	QA-001 to QA-006	6	5 PASS, 1 PASS WITH NOTE (missing values documented as a source limitation)
KPI and Country KPI	QA-007 to QA-020	14	14 PASS
Dashboard	QA-021 to QA-027	7	5 PASS, 1 PASS WITH NOTE (scrollbars and Null categories remain), 1 MANUAL (QA-025 dashboard actions)
Documentation	QA-028 to QA-029	2	2 PASS
Delivery	QA-030	1	1 MANUAL (final .twbx to be attached with the deliverables)
Total		30	26 PASS, 2 PASS WITH NOTE, 2 MANUAL

11.3 KPI Validation

KPI	Dashboard	Independent Recalculation
Overall Score	52.50	52.5008
Academic Reputation	20.29	20.2867
Research Impact	37.22	37.2192
Faculty Student Ratio	19.07	19.0717
International Students %	9.95%	9.9487%
Research Score	23.38	23.3824
Citation Score	42.98	42.9805

KPI	Dashboard	Independent Recalculation
Research Productivity	50.03	50.0282
Total Students	92.41M	92,405,187
Average Literacy	87.97	87.9667
Average Tertiary Enrollment	44.33	44.3255
Total Countries / Education Countries	136 / 119	136 / 119

Differences are due to rounding in Tableau. Accuracy was confirmed by recomputation of the displayed KPI values; no percentage accuracy claim is made beyond those checks.

11.4 Final Manual Verification

The project owner manually verified navigation, the Year filter, the University and Country filters, and dashboard linking in Tableau, and all were recorded as working. No major broken visual was found in the final dashboard screenshots.

12. Project Structure and Delivery

12.1 Repository Structure

The repository is organised by milestone, with a working folder that holds the working copy of the data, notebooks, and documentation.

Folder	Contents
Milestone 1 / Module 1	01_data_loading.ipynb and the raw data files (qs_2025_raw.csv, the_2024_raw.csv, wur_2023_raw.csv, world_bank_education_subset.csv)
Milestone 1 / Module 2	02_data_cleaning.ipynb, 03_data_standardization.ipynb, cleaned *_clean.csv files, and Normalized/ (dim_country.csv, dim_university.csv, matches_for_manual_review.csv, world_bank_matched.csv)
Milestone 2 / Module 3	04_kpi_engineering.ipynb, 05_validation.ipynb, eduvision_final_dataset.xlsx, and star_data_model/ (dimension and fact CSV files)
Milestone 2 / Module 4	dashboard_storyboard.pdf and prototype.twb
Milestone 3 / Module 5	1_University Overview.pdf and 2_Research Analytics.pdf
Milestone 3 / Module 6	3_Student Analytics.pdf and 4_Country Comparison.pdf
Milestone 3 / Final Integrated Dashboard	EduVision_Dashboard.twbx (packaged Tableau workbook)
Milestone 4 / Module 7 / Deliverables	Dashboard_Testing_Report.pdf and QA_Checklist.pdf
Milestone 4 / Module 8 / Deliverables	GitHub Repository, Final Documentation, Tableau Workbook
working/	data/ (raw, cleaned, final), docs/ (validation report, matching methodology, final validation checklist), and notebooks/ 01 to 05

12.2 GitHub Repository

The project is published on GitHub, on the Vepanjeri_Hasini branch of the course repository:

https://github.com/springboardmentor1234r/Higher-Education-Intelligence-System-for-Institutional-Performance-and-Academic-Excellence-Analysis/tree/Vepanjeri_Hasini

The repository contains a top-level README.md, milestone README files for Milestones 1 and 2, and the folders described in Section 12.1.

12.3 Module 8 Deliverables

The Milestone 4 / Module 8 / Deliverables folder contains the three Module 8 items:

1. GitHub Repository.
2. Final Documentation.
3. Tableau Workbook (EduVision_Dashboard.twbx).

12.4 Final Deliverables Checklist

Deliverable	Milestone
Raw data, cleaned data, and data loading and cleaning notebooks	Milestone 1
Star data model, eduvision_final_dataset.xlsx, KPI engineering and validation notebooks	Milestone 2
dashboard_storyboard.pdf and prototype.twb	Milestone 2
University Overview and Research Analytics dashboards	Milestone 3
Student Analytics and Country Comparison dashboards	Milestone 3
EduVision_Dashboard.twbx (integrated, all four dashboards linked)	Milestone 3
QA Checklist and Dashboard Testing Report	Milestone 4
GitHub Repository, Final Documentation, Tableau Workbook	Milestone 4

13. Milestone Summary and Tech Stack

13.1 Milestone Summary

Milestone	Focus Area	Key Outcome
1	Data Collection and Preparation	Four sources loaded, cleaned, standardised; dimension tables and matching files produced
2	KPI Engineering and Dashboard Planning	Star model with four fact tables, six KPIs, validation, storyboard, and prototype
3	Dashboard Development	Four dashboards built and integrated with navigation and filters
4	Testing and Delivery	30 QA test cases, testing report, final documentation, and GitHub delivery

13.2 Tech Stack

Area	Tools / Libraries
Data sources	QS World University Rankings 2025, THE World University Rankings 2024, World University Rankings 2023, World Bank EdStats
Data processing and KPI engineering	Python, pandas, NumPy, openpyxl, Jupyter notebooks
Visualisation	Tableau
Version control and delivery	Git and GitHub
Documentation	Markdown, Microsoft Word

14. Appendix A: Worksheet Inventory

Worksheets visible on each final dashboard.

A.1 University Overview

- UO-KPI Global Rank, UO-KPI Overall Score, UO-KPI Academic Reputation, UO-KPI Research Impact, UO-KPI Faculty Student Ratio, UO-KPI International Students
- UO-Top Universities, UO-Universities by Region, UO-Performance Trend, UO-Academic Reputation

A.2 Research Analytics

- RA-KPI Research Score, RA-KPI Citation Score, RA-KPI Research Impact, RA-KPI Research Productivity
- RA-Research Impact Comparison, RA-Top Research Institutions, RA-Research Score Analysis, RA-Research Productivity Trends, RA-Citation Performance

A.3 Student Analytics

- SA-KPI Total Students, SA-KPI International Students %, SA-KPI Faculty Student Ratio, SA-KPI International Student Score, SA-KPI Faculty Student Score
- SA-Student Diversity Trends, SA-International Student Analysis, SA-Faculty-to-Student Ratio Analysis, SA-Enrollment Comparisons, SA-Student Distribution Analysis

A.4 Country Comparison

- CC-KPI Total Countries, CC-KPI Education Countries, CC-KPI Avg Literacy, CC-KPI Avg Tertiary Enrollment
- CC-Regional Education Comparison, CC-Country Ranking Comparison, CC-Education Performance Benchmarking, CC-Top Performing Countries

15. Appendix B: Glossary

Term	Meaning
Grain	The level of detail one row in a table represents, for example one row per university per source per year.
Dimension table	A table of descriptive attributes used to group or filter, such as dim_university and dim_country.
Fact table	A table of measurements linked to dimensions, such as fact_research.
Star schema	A data model with fact tables connected to surrounding dimension tables.
Composite key	A key made of several columns that together identify a row: university_id + year + source.
Referential integrity	Every identifier used in a fact table exists in the matching dimension table, with none orphaned or invented.
Min-max scaling	Rescaling values to a 0 to 100 range using the lowest and highest values in a group.
Percentile rank	The percentage of values in a group that fall at or below a given value.
True percentage vs score	A true percentage is a real share of students; a QS score is a 0 to 100 rating and is not a percentage.
Parameter / filter	Controls that change what a Tableau worksheet shows.